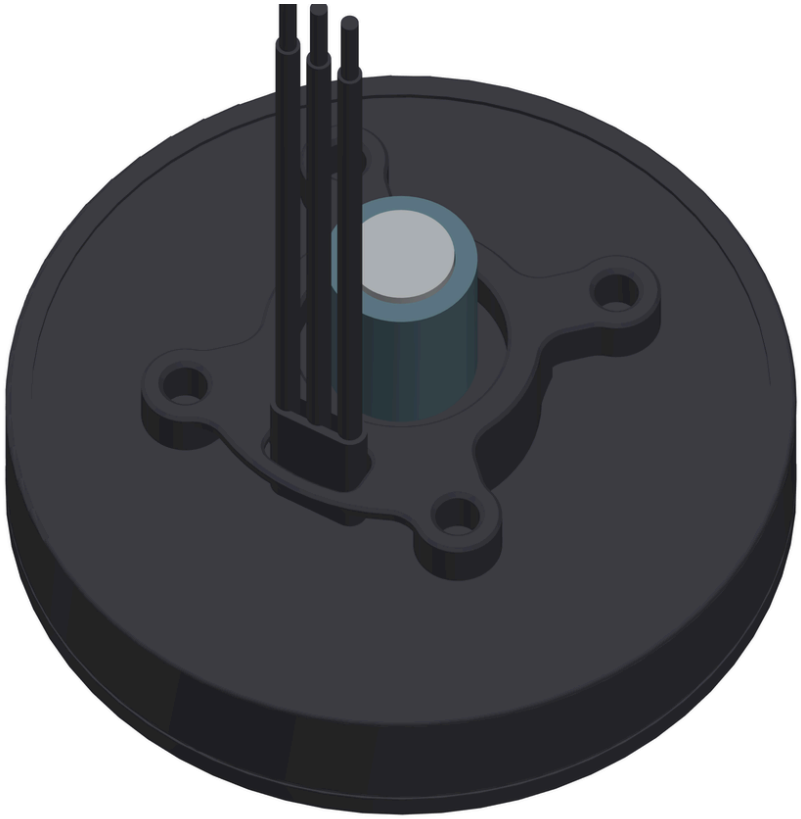
Magnets

How they work, how to fit them.



Designed by Athanasia Nikolova

All data come from tests and many real builds, not only from theory. v1.2 · November 2026

Teacher only



One thing per page

One action, one big picture. Short sentences, much white space.



Color is never the only clue

Every color also has a number, a letter or a name next to it. The part colors stay distinct with color-vision deficiency (Okabe-Ito palette).



Clear print

Instructions at 14 pt, everything else at least 12 pt, in Atkinson Hyperlegible. The text has a contrast of at least 4.5 : 1 to the page (WCAG 2.x, AA).



For a screen reader

The PDF is tagged (PDF/UA-1): headings, lists and reading order are marked, and every picture has a text alternative that is read aloud.



Print it right

In color, at 100 % - not 'fit to page'.



Pace

One page, then the action, then the next page. For a student who tires quickly: one lesson per session. The check at the end of each lesson is a good place to stop.

Safety first



Grown-up nearby

Work with your teacher or mentor in the room.



Fingers

Strong magnets snap together: keep them apart, they pinch fingers.



Glue

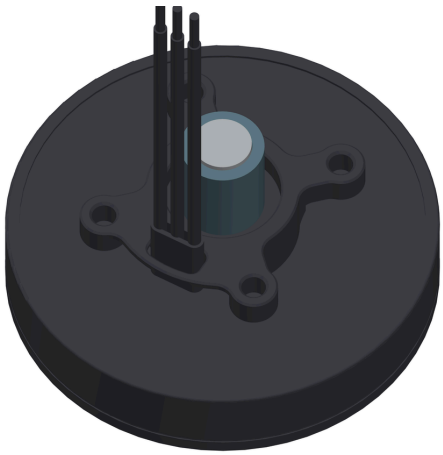
Super glue sticks skin in seconds. Use a toothpick, not your finger.



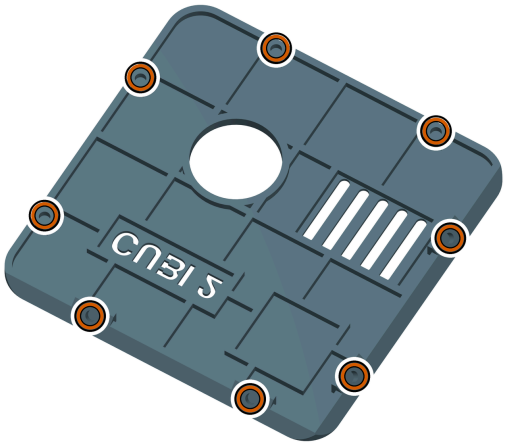
Small children

Keep the magnets away from small children - a swallowed magnet is dangerous.

Two kinds of magnets



On each motor · ×3



On the covers · ×48

On each motor · ×3

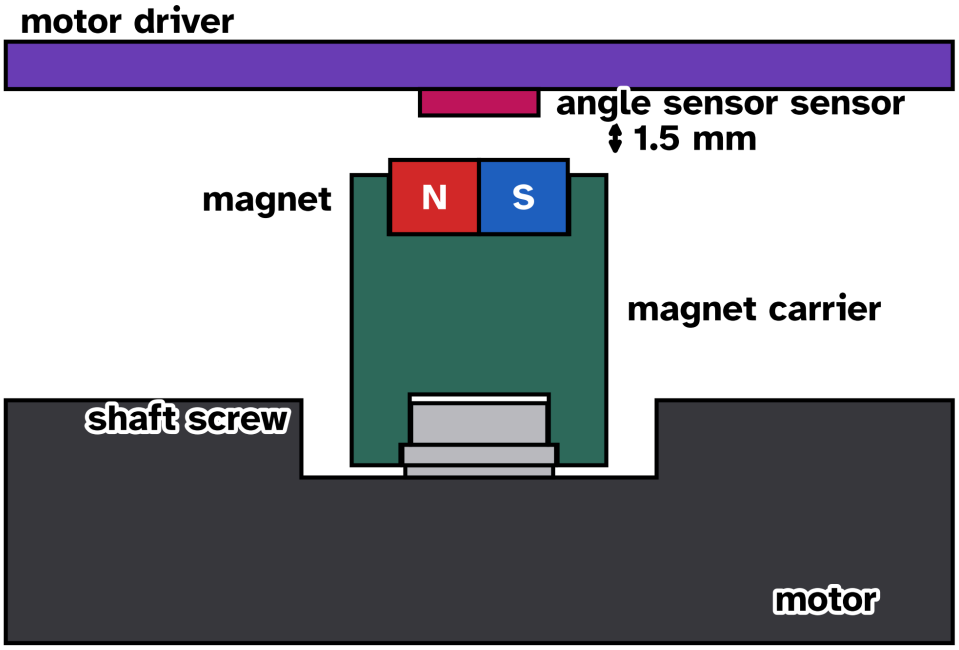
A round magnet Ø6 x 2.5 mm on the end of the motor shaft. It turns with the wheel. The motor driver reads it and knows the exact angle of the motor.

On the covers · ×48

Small magnets Ø3 x 2 mm. They hold the 3 covers on the box - no screws. Each cover magnet sits opposite a magnet in the box or in a beam.

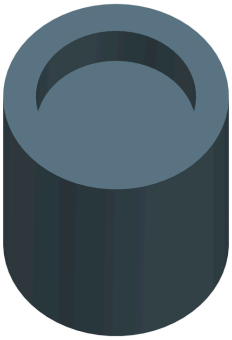
How the motor knows its angle

The magnet has a North half and a South half. The sensor under it feels where they point.

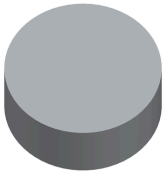


- The magnet must sit in the middle of the shaft - the handler keeps it there.
- The gap to the sensor is 1.5 mm. Nothing may touch.

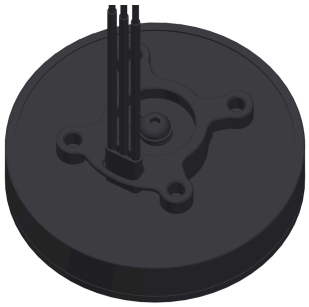
Motor magnets: you need



3 magnet carriers



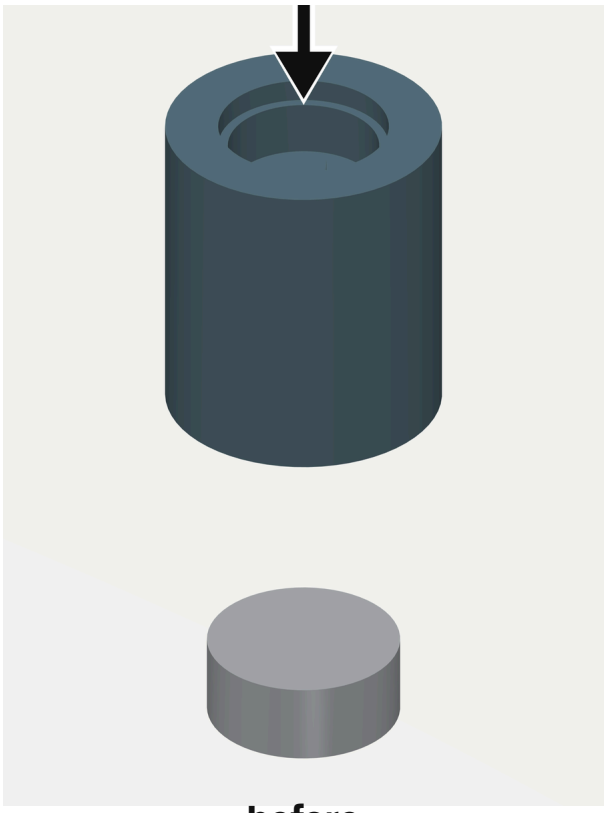
3 magnets Ø6 x 2.5



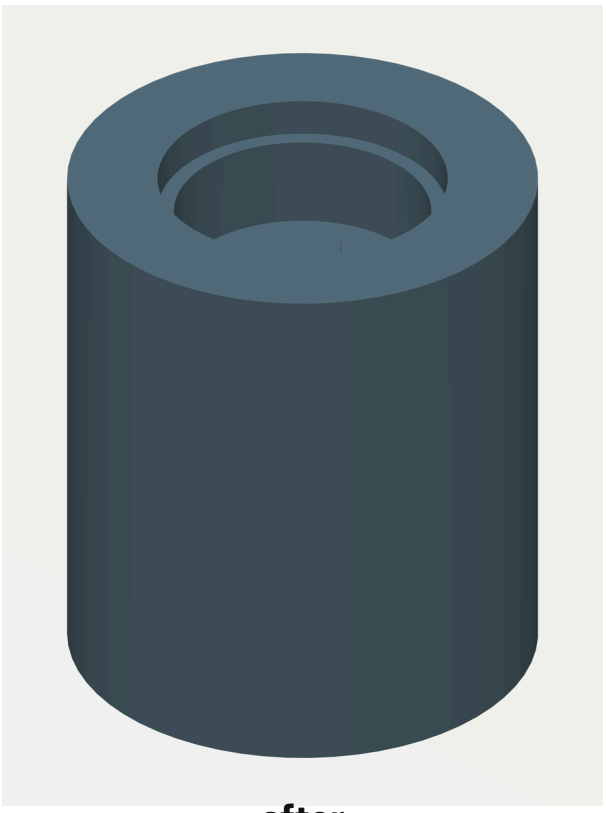
the 3 motors

- The magnet-handler (printed tool, magnet-handler.step)
- A hard, flat table
- Super glue (CA), a little - only for the carrier on the motor

1 · Magnet into the carrier



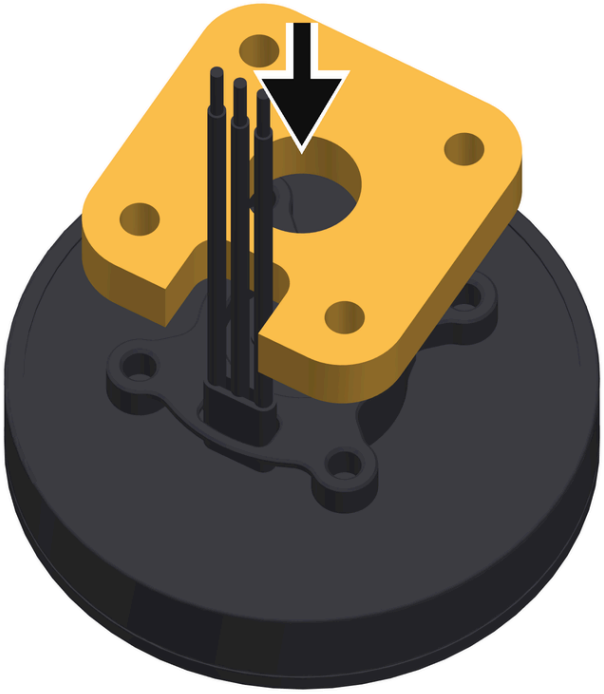
before



after

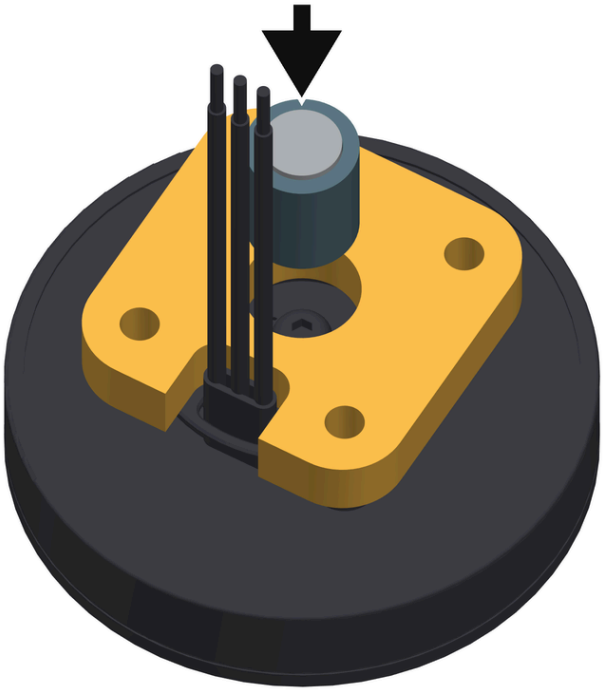
- Put the magnet flat on the table.
- Hold the carrier upside down over it, the round hole down.
- Press hard, straight down, until it stops. No glue.

2 · Screw the handler on



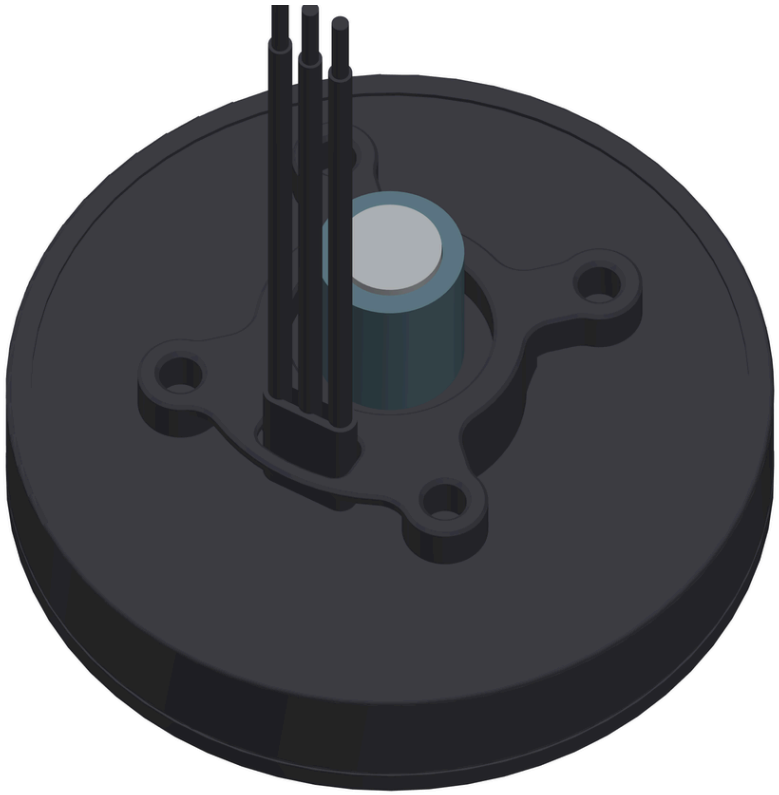
- Screw the magnet-handler onto the back of the motor.
- 4 M3 screws in the motor's own 4 holes. Its slot goes over the 3 motor leads.

3 · Glue and press



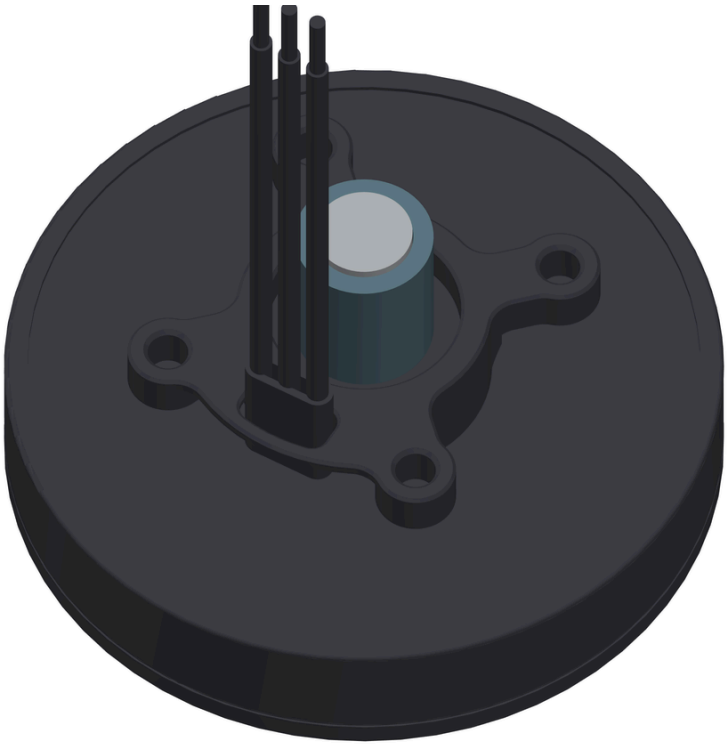
- A small drop of glue on the screw at the back of the motor.
- Push the carrier through the handler until it sits on the screw.
- Its little hex pin goes into the screw - do not turn it.

4 · Wait, then take the handler off



- Leave it a few minutes: the handler keeps it straight while the glue sets.
- Unscrew the handler. The carrier stays, centered.

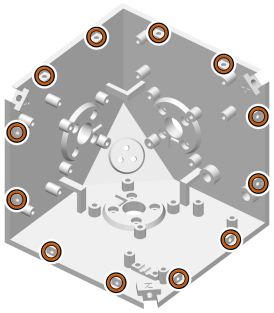
Check the motor magnet



- The carrier stands straight, the magnet is flat on top.
- Turn the wheel by hand: the magnet turns round, it does not wobble.
- Three motors - three magnets.

Cover magnets: where

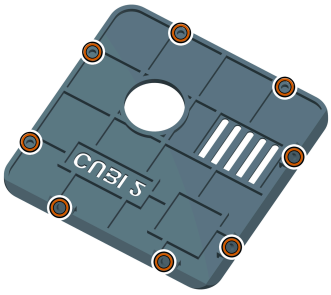
48 magnets, always in pairs.



In the box



In each tie beam



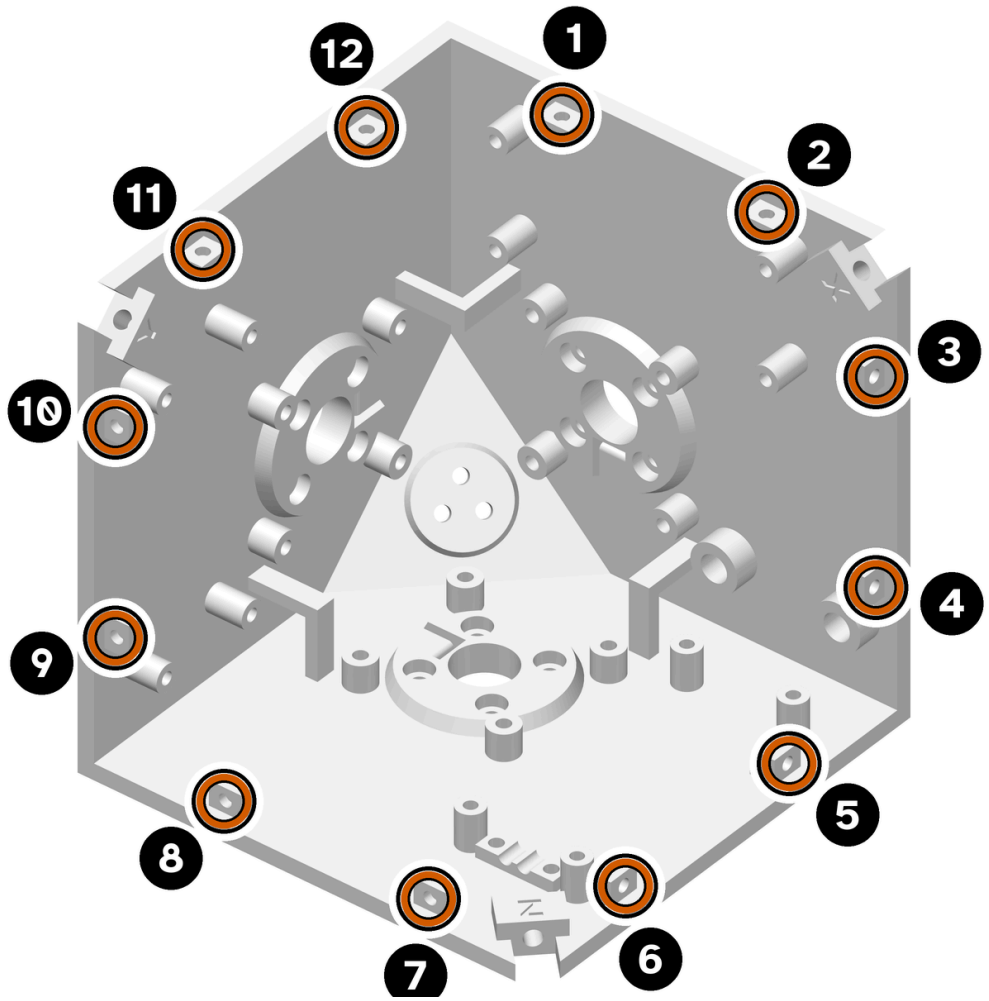
In each cover

Where	How	Magnets
In each cover	8 × 3 covers	**24**
In the box	12 (4 on each open face)	**12**
In each tie beam	4 × 3 beams	**12**
All together		**48**

- First all the box and beam magnets. Then the covers, one at a time.
- You need: super glue, a toothpick, a marker pen.

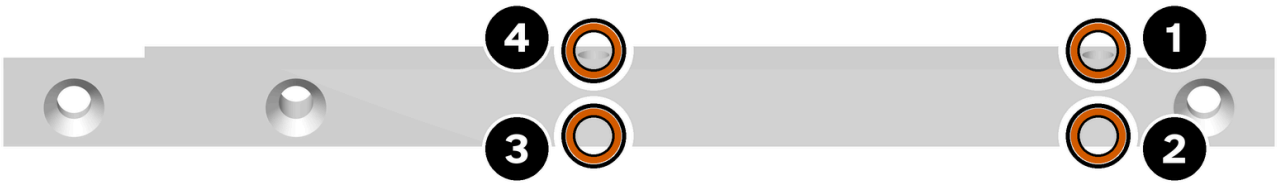
In the box · 12

4 small bases on the top of each open face.



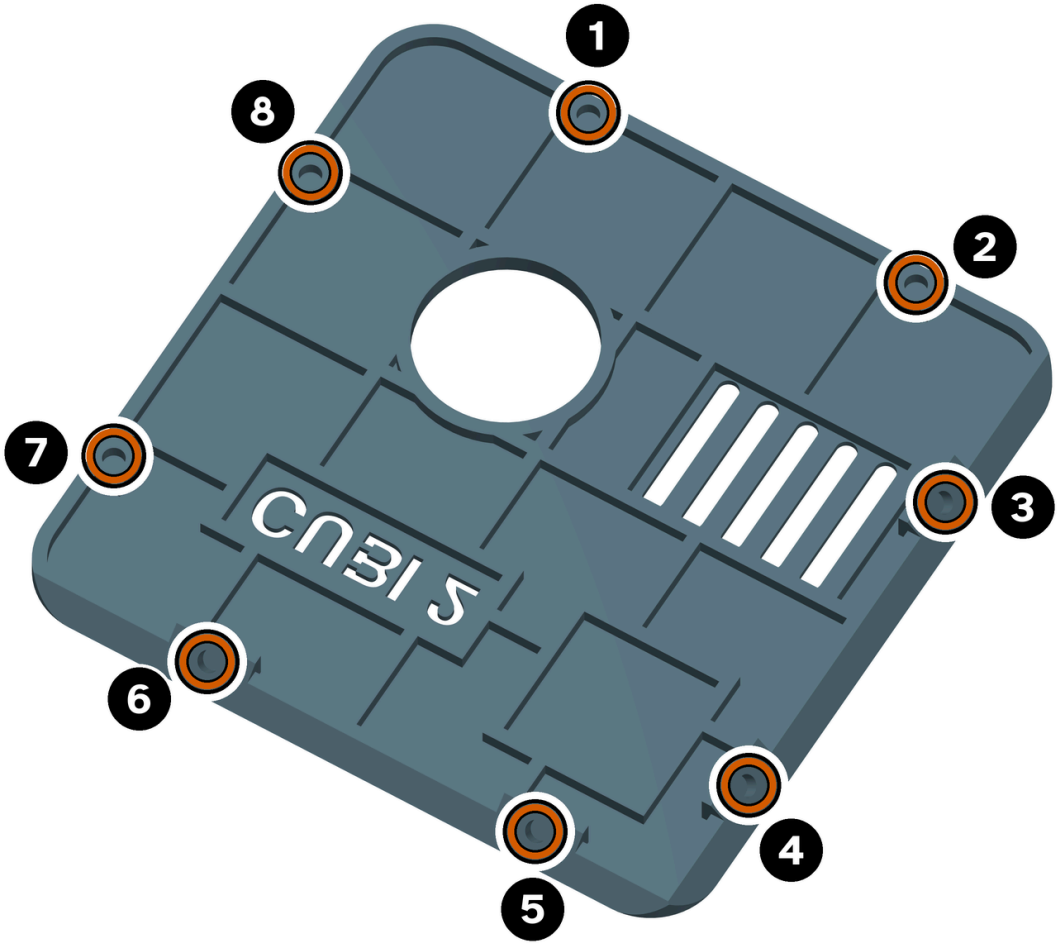
In each tie beam · 4

The 3 beams with the tie hole.



In each cover · 8

Inside face of the cover.

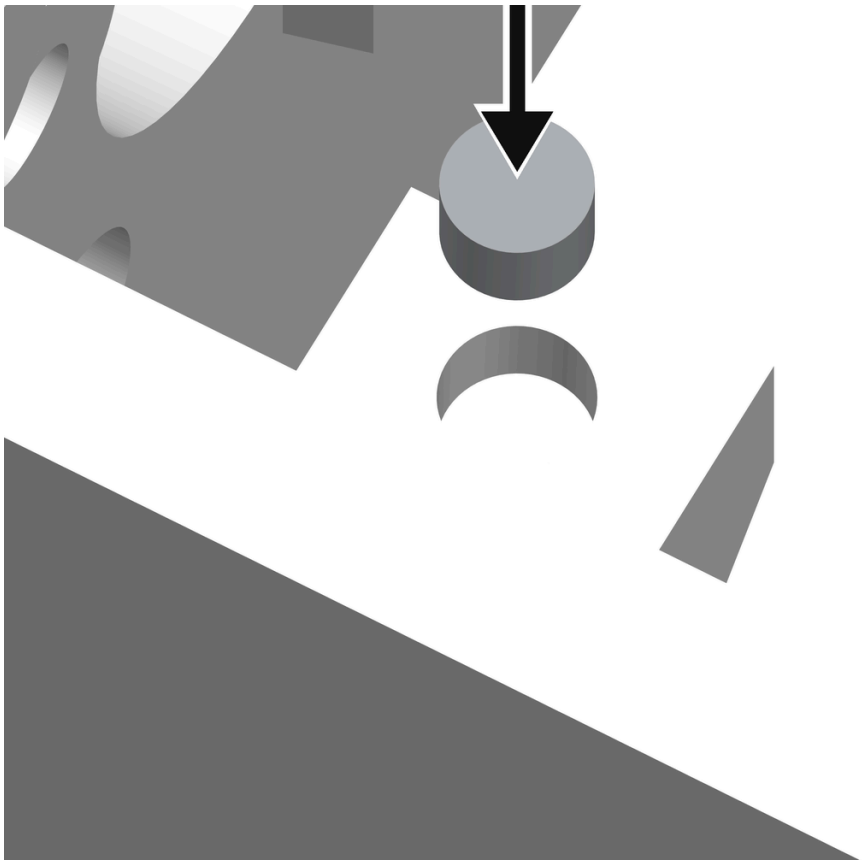


1 · Glue in the pocket



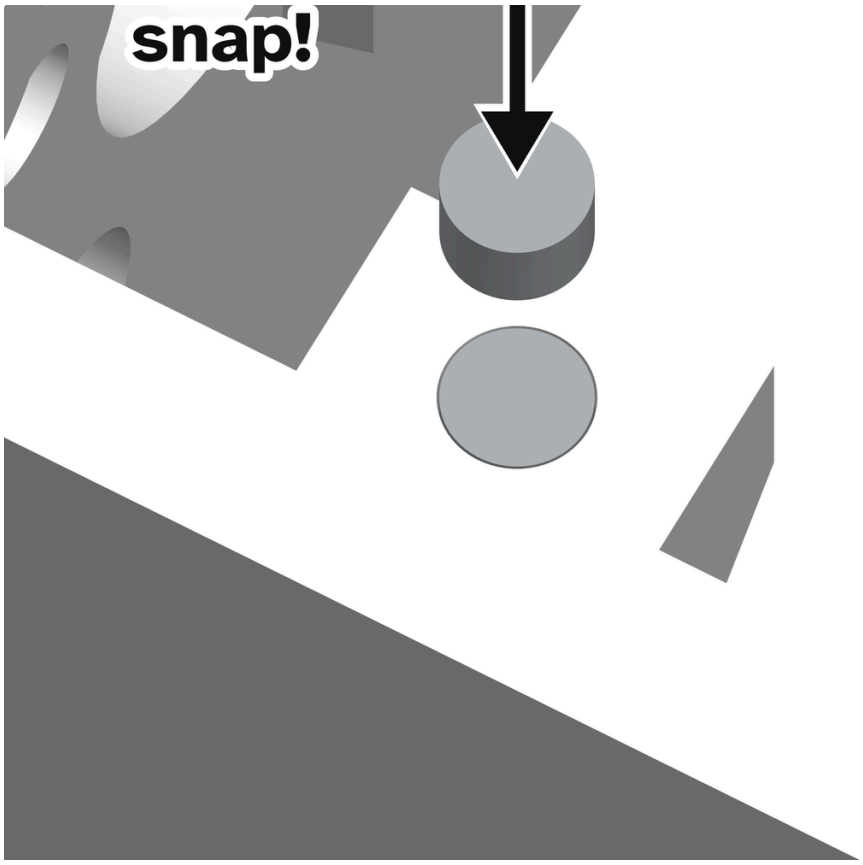
- A tiny drop of glue in the pocket, with a toothpick.
- Box and beams first - every pocket.

2 · Magnet in, flat



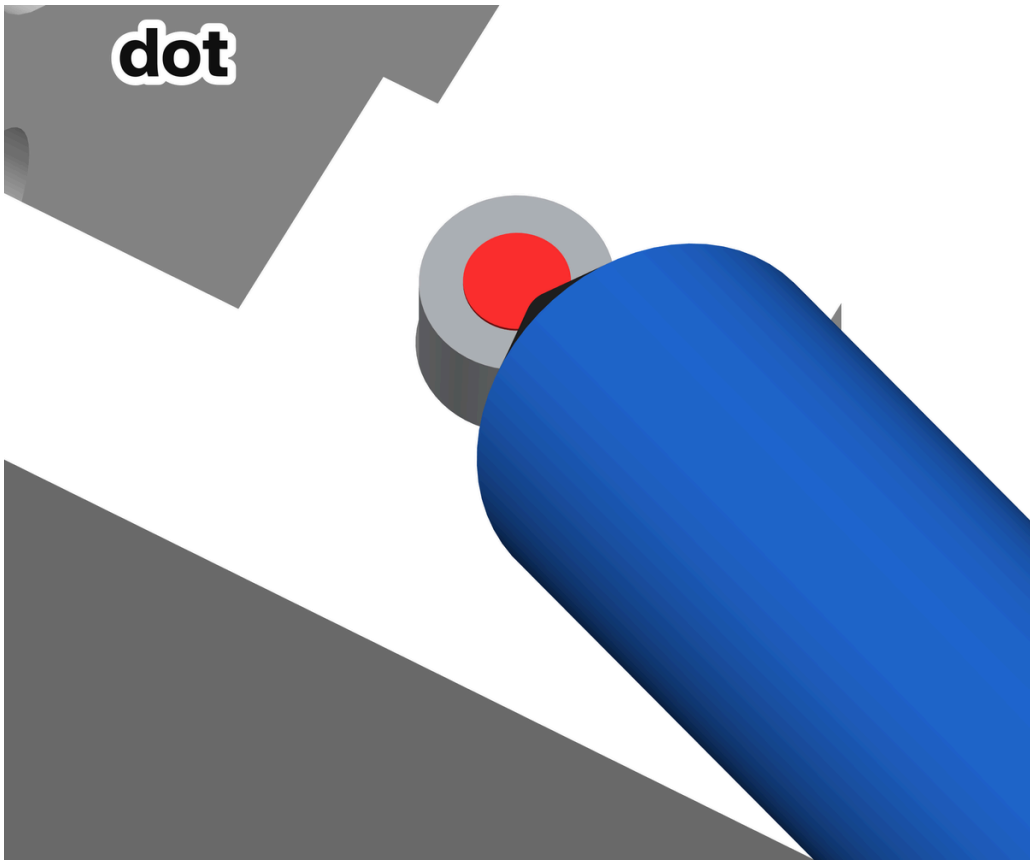
- Push the magnet in with a plastic or wooden stick - a steel tool pulls it out.
- Flat with the surface. Any way round here.

3 · Let it snap



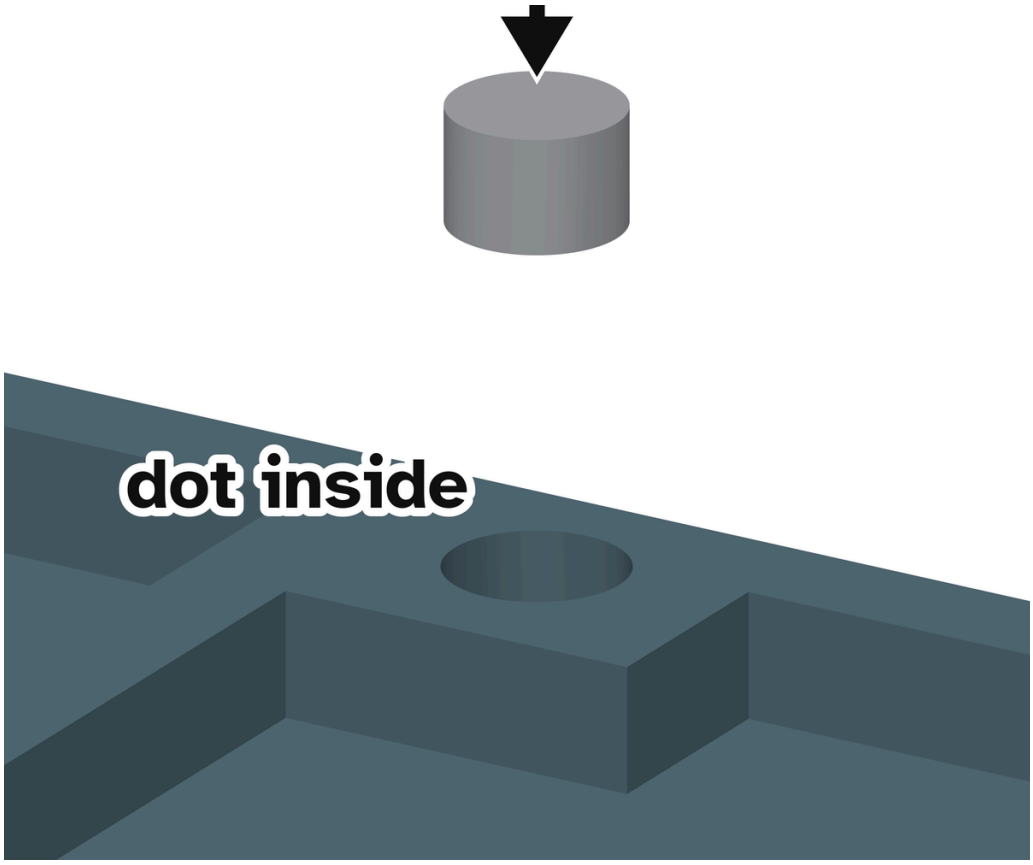
- Put a loose magnet on top of a glued one.
- It snaps on and turns itself the right way.

4 · Mark its top



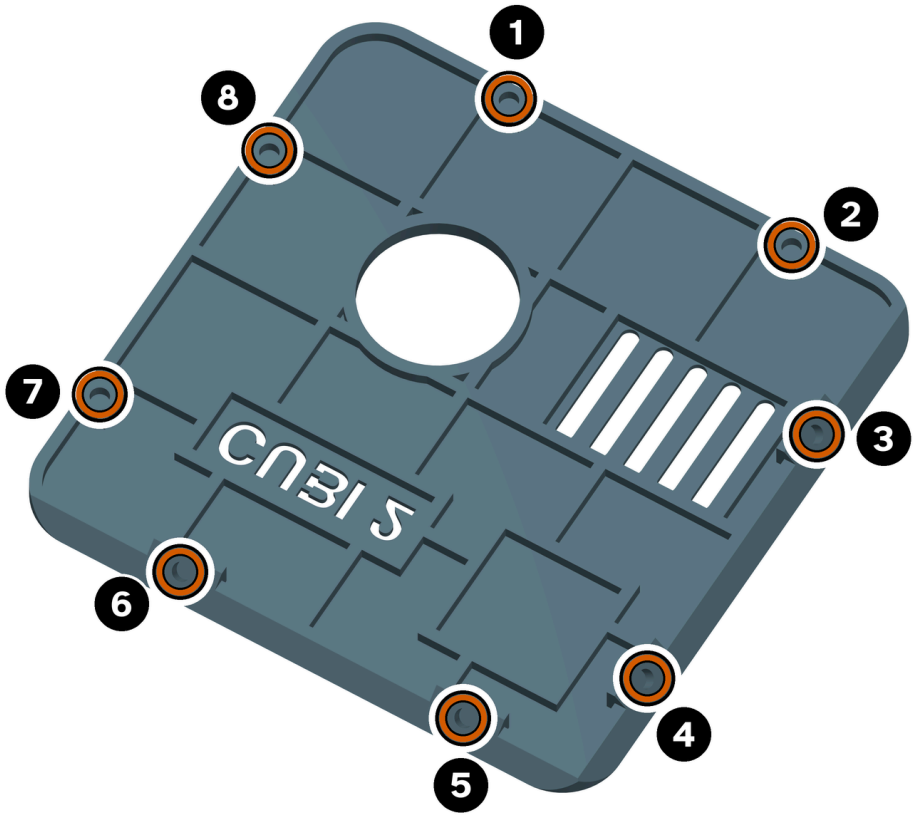
- A dot with the marker on its top face.
- Pull it off. The dot shows which face goes into the cover.

5 · Into the cover - dot first



- A tiny drop of glue in the cover pocket.
- Push the magnet in with the DOT INSIDE, flat with the surface.
- Do the 8 pockets of this cover, each from its own partner.

6 · Close the cover



- Wait for the glue, then put the cover on its face: it jumps into place.
- Write the face letter (X, Y or Z) inside the cover: it belongs there.

Good to know



Glue only on the pocket

Never on the face that touches another magnet.



Test each pair

Before the glue sets: the cover jumps into place, it is not pushed away.